

Model checking and comparison

Bert van der Veen

University of Bayreuth

Model checking

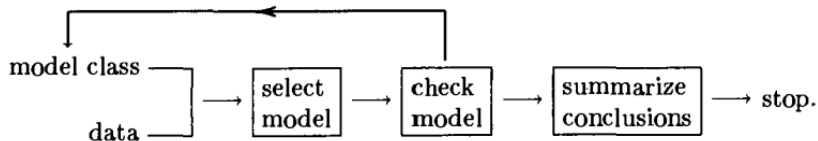


Figure 1: McCullagh and Nelder (1989) workflow

VGLMM assumptions

We have made various assumptions in the models we fitted so far:

1. Species responses are independent (violated $\rightarrow$ JSMD, tomorrow)
2. Same dispersion for all species (violated $\rightarrow$ vector GLM)
3. Correct distribution and link function
4. Correctly specified model structure (species-specific vs. common effects)
5. No outliers

Mean-variance

Methods in Ecology and Evolution



Methods in Ecology and Evolution 2017, **8**, 1408–1414

doi: 10.1111/2041-210X.12843

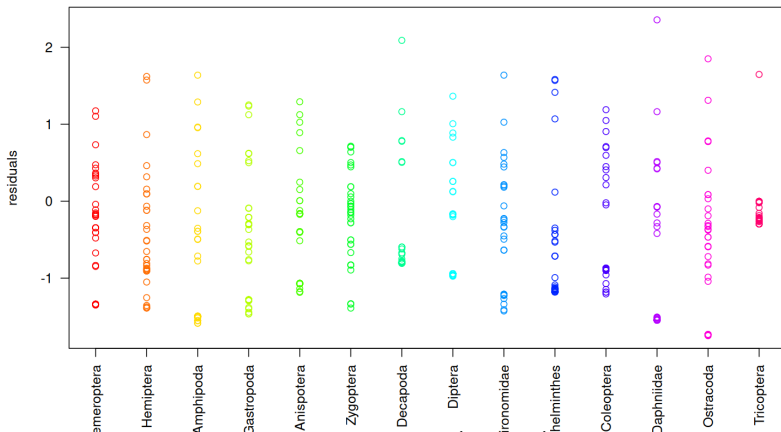
FORUM

The central role of mean-variance relationships in the analysis of multivariate abundance data: a response to Roberts (2017)

David I. Warton^{*,1} and Francis K. C. Hui²

The mean-variance relationship has been one of the main drivers of model-based multivariate methods.

Wetlands: species dispersion



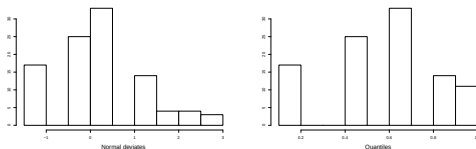
Clearly, not all these species have the same (residual) dispersion.

Randomized Quantile residual (Dunn and Smyth 1996)

- ▶ Gold standard residual
- ▶ Better suited for small samples and discrete data types
- ▶ Exactly normally distributed
- ▶ Suitable for all kinds of models

Continuous

$$r_Q = \Phi^{-1} \left\{ \mathcal{F} \left(y_i; \hat{\mu}_i, \hat{\phi} \right) \right\} \quad (1)$$



(V)GLMM checking



(V)GLMM residuals

Random effect estimates $\hat{\mathbf{u}}$ have a very similar relation to the random effect $\mathbf{u}$ as the residual $\hat{\epsilon}$ does for the error ϵ .

Conditional

$$g\{\mathbb{E}(y_{ij}|\mathbf{x}_i)\} = \mathbf{x}_i^\top \hat{\boldsymbol{\beta}}_j + \mathbf{z}_i^\top \hat{\mathbf{u}}_j \quad (2)$$

Unconditional

$$g\{\mathbb{E}(y_{ij}|\mathbf{x}_i)\} = \mathbf{x}_i^\top \hat{\boldsymbol{\beta}}_j \quad (3)$$

How do we calculate the residual?

- ▶ should we condition on the estimate of the random effect?
- ▶ simulate from conditional distribution?

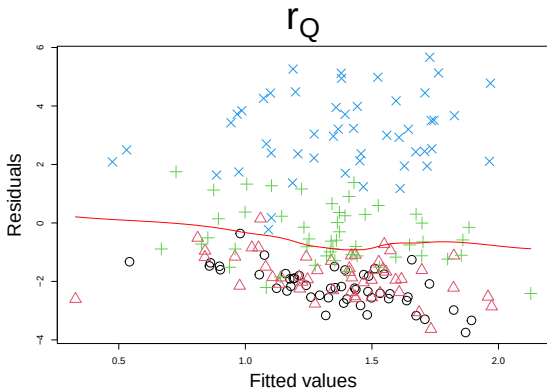
i.e. a range of options

Simulation: grouping of errors

```
n <- 200
ngroups <- 4
alpha <- 0.5
beta <- -1
x <- rnorm(n, sd = 0.2)

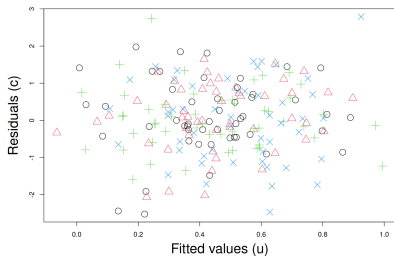
fac<-rep(1:ngroups,each=n/ngroups)
e <- seq(from=-2,to=2,length.out=ngroups)
mu <- exp(alpha + beta*x + e[fac])
y <- rpois(n = n, lambda = mu)
```

Example: Poisson residuals (grouping)

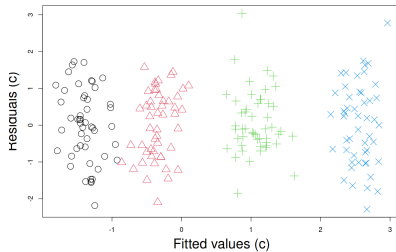


Residual diagnostics: Poisson residuals (grouping)

Conditional residual - Unconditional fitted



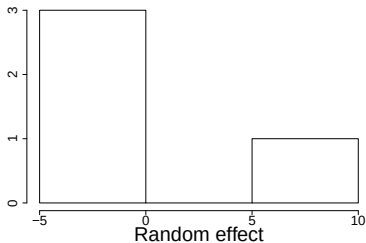
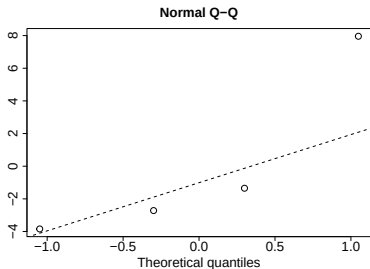
Conditional residual - conditional fitted



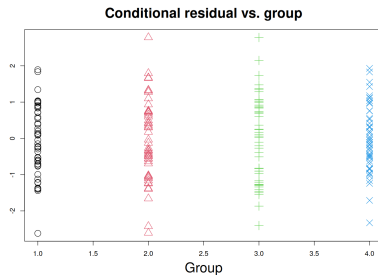
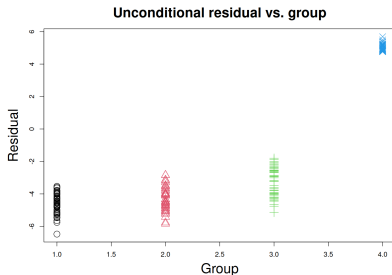
GLMM: checking random effect assumptions

- ▶ random effect is a type of residual
- ▶ $\hat{u}_j$ is an estimate of the mean or mode of $p(u_j|y_i)$
- ▶ we treat $\hat{u}_j$ as a sample of the random effect distribution
- ▶ so we check assumptions (marginal normality, constant variance, independence, no outliers)!
- ▶ difficult with small number of groups
- ▶ needs to be done for every random effect

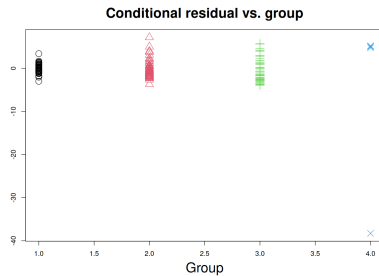
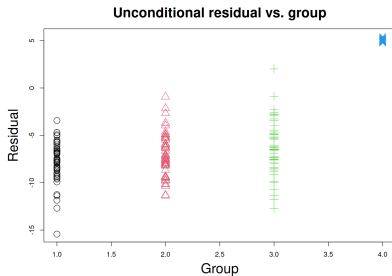
GLMM diagnostics: outlier in random effects



GLMM diagnostics: outlier (residuals vs. group)



GLMM diagnostics: unequal variance



What makes a good statistical model?

One that helps to answer your research question:

- ▶ Accurately represents the data generating process
- ▶ Not too difficult to interpret
- ▶ Is robust

What makes a good statistical model?

One that helps to answer your research question:

- ▶ Accurately represents the data generating process
- ▶ Not too difficult to interpret
- ▶ Is robust

Principle of parsimony: if two competing models fit the data equally well, prefer the simpler one.

A more complex model will always fit better, but it may overfit. Complexity must be justified.

Omitted variable bias

Occurs when we omit a variable that truly matters, i.e., we have:

$$g\{\mathbb{E}(y_i|x_{i1}, x_{i2})\} = \alpha + x_{i1}\beta_1 + x_{i2}\beta_2 \quad (4)$$

but fit without x_{i2} . The consequence:

$$y_i = \alpha + x_{i1}\beta_1 + \epsilon_i, \quad \text{with } \epsilon_i = x_{i2}\beta_2 + \epsilon_i^* \quad (5)$$

- ▶ ϵ_i may be correlated with x_{i1}
- ▶ Causes **biased** parameter estimates and incorrect standard errors
- ▶ Not including a confounding variable is worse than including it

Likelihood ratio test

Is improved fit due to noise or is the alternative model actually better?

Procedure

- ▶ Fit two models: M_0 with k parameters and M_1 with r
- ▶ Calculate likelihood ratio $\Lambda = \log \left(\frac{\mathcal{L}(\mathbf{y}; \Theta_0)_{M_0}}{\mathcal{L}(\mathbf{y}; \Theta_1)_{M_1}} \right)$
- ▶ $\mathcal{L}(\mathbf{y}; \Theta_0)_{M_0} \leq \mathcal{L}(\mathbf{y}; \Theta_1)_{M_1}$
- ▶ $-2\Lambda \sim \chi^2(k_1 - k_0)$ under the null
- ▶ $p \geq 0.05$ difference in likelihood is due to sampling

LRT approximation assumptions

- ▶ $n \rightarrow \infty$
- ▶ Θ_0 contained in Θ_1 : nested models
- ▶ The true parameter is in the interior of the parameter space
- ▶ Model is “identifiable”
- ▶ Hessian matrix is sufficiently close to the Fisher information
- ▶ y_i are independent

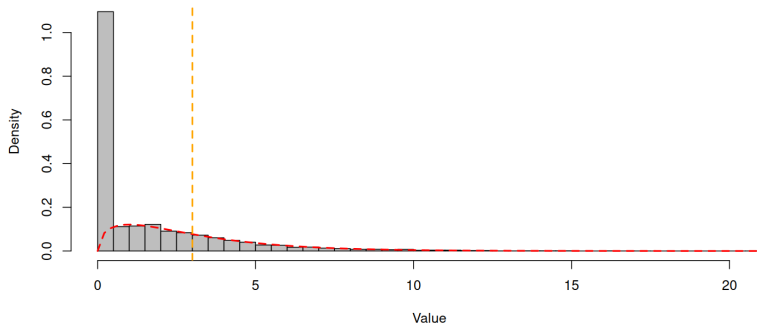
These assumptions may fail, especially in models more complex than VGLMs

Alternatively: LRT by simulation.

Boundary issues

Variances σ^2 are positive only. If we compare to a model without random effect, it is like saying $\sigma^2 = 0$: the null is “on the boundary”, and the sampling distribution has a spike at zero.

Mixture of Chi-squared Distribution with Mass at 0 and 3



Wald-statistic

The wald-statistic is reported in summary:

$$W = (\hat{\theta} - \theta)^2 / \text{var}(\hat{\theta}) \quad (6)$$

we test if the parameter estimate is different from zero. For this, we make the following assumptions:

- ▶ We have a good estimate of the standard error
- ▶ The estimator is normally distributed (large samples)
- ▶ The estimator is centered around the true parameter

Information criteria

A different paradigm:

Find the best model amongst a set of models.

Best:

- ▶ Penalise complexity (number of parameters)
- ▶ By fit (likelihood)

Most commonly:

- 1) AIC: Akaike's Information Criterion (Akaike 1974)
- 2) BIC: Bayesian Information Criterion (Schwarz 1978)

Lower = better

Akaike's Information Criterion

$$\text{AIC} = -2\mathcal{L}(\mathbf{y}; \Theta) + 2k \quad (7)$$

- ▶ Penalizes model complexity
- ▶ (approximately) Measures information loss to the true data generating process
- ▶ Asymptotically

AIC tends to select too complex models with little data. Finite sample correction (Sugiura 1978):

$$\text{AIC}_c = \text{AIC} + \frac{2k(k+1)}{n-k-1} \quad (8)$$

Find the model that predicts best

Bayesian Information Criterion

$$\text{BIC} = -2\mathcal{L}(\mathbf{y}; \Theta) + k \log(n) \quad (9)$$

So the penalty is different.

Find the model closest to the "true" model

Connection of AIC and LRT

Rule of thumb: difference of 2 points means a model is better

$$\begin{aligned}
 \Delta\text{AIC} &= \text{AIC}_{M_1} - \text{AIC}_{M_0} \\
 &= 2\mathcal{L}(\mathbf{y}; \Theta_0) - 2\mathcal{L}(\mathbf{y}; \Theta_1) + 2k_1 - 2k_0 \quad (10) \\
 &= -2\Lambda + 2(k_1 - k_0)
 \end{aligned}$$

So AIC with a rule of $\delta = 2$ can be seen as a more liberal LRT (Sutherland et al. 2023)

The cult of (A)IC

Presentation by Mark Brewer

"Always use (A)IC for model comparison"

My perspectives

- ▶ Use common sense
- ▶ Do not blindly test all models ("dredging"): by chance, unrelated predictors will be selected
- ▶ Use model comparison techniques in moderation



Don't take the "best" model paradigm too seriously

Example 2: Swiss birds

We had this model:

```
model3 <- gllvm(y, X = X, formula = ~(bio_1+bio_12|1),
               family = "binomial", num.lv = 0, beta0com = TRUE)
```

Let's compare it to:

```
model4 <- gllvm(y, X = X, formula = ~(bio_12|1),
               row.eff = ~bio_1+bio_12, studyDesign = X,
               family = "binomial", num.lv = 0, beta0com = TRUE)
```

and perhaps

```
model5 <- gllvm(y, X = X, formula = ~(bio_12|1),
               family = "binomial", num.lv = 0, beta0com = TRUE)
```

Example 2: the models

The ecological assumptions of the models are a little different:

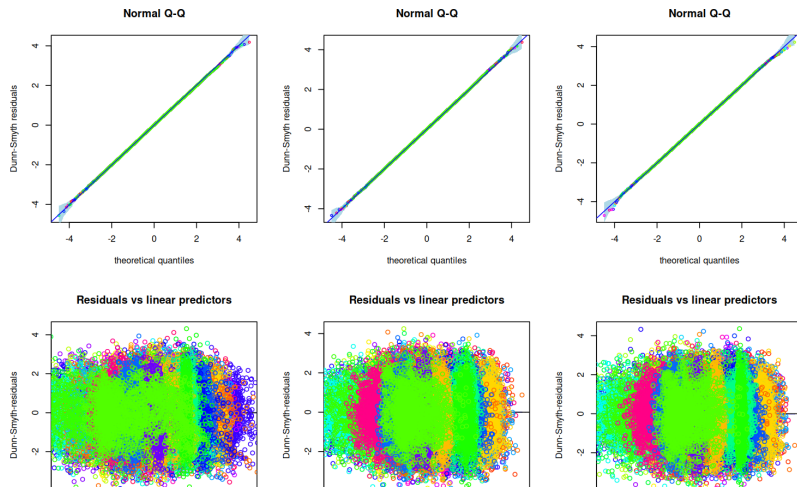
- 1) The first model assumes correlated species-specific responses to temperature and precipitation
- 2) The second model assumes a common mean temperature effect, with species-specific variation in precipitation
- 3) The third model assumes no responses to temperature at all

So complexity is 1) > 2) > 3)

The models are nested, so we can use a hypothesis test, or information criteria, for comparison.

Example 2: residuals

The residuals of all three look OK:



Example 2: comparison

```
anova(model4, model5) # model with mean temperature effect is "better"
```

```
## Model 1 : y ~ NULL
## Model 2 : y ~ NULL
```

##	Resid.Df	D	Df.diff	P.value
## 1	141339	0.0000	0	
## 2	141338	395.4487	1	0

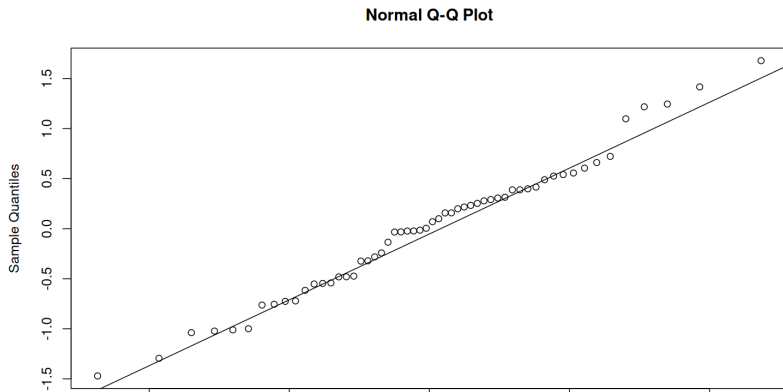
```
anova(model3, model4) # Model with temperature REs is "better"; note null on the boundary
```

```
## Model 1 : y ~ NULL
## Model 2 : y ~ NULL
```

##	Resid.Df	D	Df.diff	P.value
## 1	141338	0.00	0	
## 2	141335	14208.13	3	0

Example 2: examining the REs

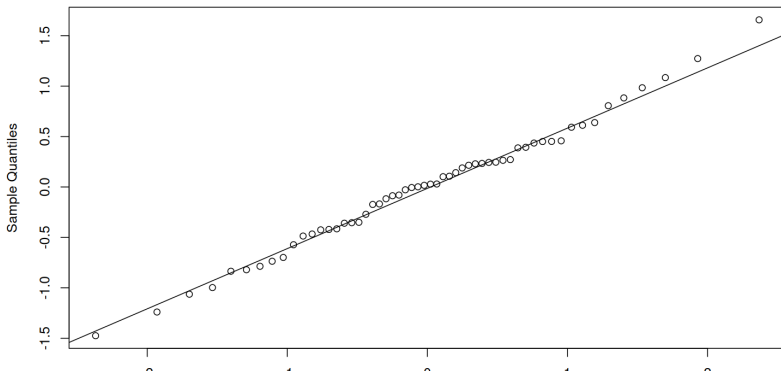
```
qqnorm(coef(model13, "Br")["bio_1",])
qqline(coef(model13, "Br")["bio_1",])
```



Example 2: adjusting the model

```
model6 <- gltvm(y, X = X, formula = ~(bio_1+bio_12+slp|1), family = "binomial", num.lv = 0, beta0com = TRUE)
qqnorm(coef(model6, "Br")["bio_1",])
qqline(coef(model6, "Br")["bio_1",])
```

Normal Q-Q Plot



Example 2: information criteria

```
AIC(model3, model6) # lower is "better"
```

```
##          df      AIC
## model3    9 124112.3
## model6   14 121342.8
```

```
AICc(model3, model6)
```

```
##          df      AICc
## model3    9 124102.3
## model6   14 121327.8
```

```
BIC(model3, model6)
```

Convergence

All of this assumes our model has converged. If the model has not converged, it may not be valid in any way.

Convergence

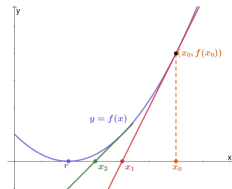
All of this assumes our model has converged. If the model has not converged, it may not be valid in any way.

How do we know we have converged?

At the MLE: gradient = 0, Hessian positive definite.

Stopping criteria:

- ▶ Gradient ≈ 0 (good)
- ▶ ℓ improvement $<$ threshold (good)
- ▶ Max iterations reached (bad)
- ▶ Hessian check fails (bad)



Convergence: what to do

- ▶ Standardise all continuous covariates before fitting
- ▶ Near-zero variance estimate: the random effect may be unnecessary; consider dropping it
- ▶ Check for warnings: gradient, Hessian, or maximum iterations reached
- ▶ Increase maximum iterations or tighten tolerance: `maxit = 10000`, `reltol = 1e-14`
- ▶ Try different starting values: `starting.val = "res"` or `"zero"`
- ▶ Try a different optimizer: `optimizer = "optim"` or `"nlminb"`
- ▶ Few levels (< 5): treat the effect as fixed instead

Checking convergence in `gllvm`

- ▶ `fit$convergence == 0`: relative convergence criterion met
- ▶ `max(abs(c(fit$TMBfn$gr())))) < 0.01`: gradients near zero
- ▶ `all(eigen(-fit$Hess$Hess.full)$values < 0)`: Hessian negative definite
- ▶ Warning "sd.errors could not be calculated, due to singular fit": Hessian not invertible

see [gllvm convergence vignette](#)

Technical caveats

- ▶ **Hypothesis testing:** Wald statistics and p -values are approximate; interpret with caution
- ▶ **Variance testing:** $H_0 : \sigma^2 = 0$ is on the boundary of the parameter space; LRT accepts alternative too often
- ▶ **Model selection:** AIC is available but treat as informal; boundary issues apply
- ▶ **Random effect diagnostics:** `qqnorm(ranef(model))` checks normality of estimated random effects

Technical caveats

- ▶ **Hypothesis testing:** Wald statistics and p -values are approximate; interpret with caution
- ▶ **Variance testing:** $H_0 : \sigma^2 = 0$ is on the boundary of the parameter space; LRT accepts alternative too often
- ▶ **Model selection:** AIC is available but treat as informal; boundary issues apply
- ▶ **Random effect diagnostics:** `qqnorm(ranef(model))` checks normality of estimated random effects

And even more so for VGLMMs

- ▶ Always check (residual) assumptions